

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Public Bus Live Tracking & Crowding Estimator

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1. Introduction

The Public Bus Live Tracking & Crowding Estimator is a smart public transportation system that provides real-time bus location, estimated arrival time (ETA), and crowding information. The system receives GPS data from buses and passenger information from ticketing sensors and processes this data to provide useful information to commuters and fleet controllers.

2. Purpose

The purpose of this system is to improve public transportation monitoring and commuter experience by providing accurate real-time bus information and helping fleet controllers monitor active buses.

3. Scope

The system will:

- Receive GPS data from active buses.
- Track the real-time location of buses.
- Estimate bus arrival times.
- Estimate crowding levels using ticketing sensor data.
- Display bus information to commuters.
- Allow fleet controllers to monitor active buses.
- Generate alerts when required monitoring thresholds are exceeded.

4. Actors

- **Commuter:** Views bus location, ETA, and crowding information.
- **Fleet Controller:** Monitors active buses and fleet conditions.

- **Bus GPS & Ticketing Sensor System:** Provides GPS and passenger-related sensor data.

5. Functional Requirements

ID	Requirement
FR-001	The system shall receive GPS data from active buses at regular intervals.
FR-002	The system shall estimate the crowding level of buses using ticketing sensor data.
FR-003	The system shall provide commuters with bus location, ETA, and crowding information.
FR-004	The system shall allow fleet controllers to monitor active buses.
FR-005	The system shall generate alerts when defined monitoring thresholds are exceeded.

6. Non-Functional Requirements

ID Requirement

- NFR-001 The system shall support up to 1,000 active transit vehicles simultaneously.
- NFR-002 The system shall provide secure communication and maintain at least 99.5% system availability.

7. System Workflow

1. Bus GPS units collect the current location of buses.
2. GPS data is sent to the GPS Data Injection Service.
3. The Bus Tracking Service processes the location information.
4. ETA Estimation and Crowding Estimation services process the required data.
5. The processed information is stored in the Transit Data Store.
6. The Commuter Information Service provides information to commuters.

7. The Fleet Control Dashboard provides monitoring information to fleet controllers.

8. Architectural Style

The system uses a **Microservices Architecture**. Major functions such as GPS data processing, bus tracking, ETA estimation, crowding estimation, commuter information, and fleet monitoring are implemented as separate services. This allows individual services to be developed, maintained, and scaled independently.

9. Expected Outcome

The system will provide real-time and useful transportation information to commuters while helping fleet controllers monitor the public bus fleet efficiently.